Standing wave sound sculpture: A compositional example

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Abstract

This paper will discuss a standing wave sound sculpture which I created at Monash University in November last year. The term standing wave sound sculpture pertains to a sculpture using the materials of sound, which are the rapid vibrations of air particles. This differs from the more common definition of sound sculpture, which is a physical object that can create, or be used to create sound.

The standing wave sound sculpture intends to create in a space, in this case a stairwell, a sound environment that not only changes in time, but also changes in the ears of the observers as they move through the space.

Just like a traditional sculpture that changes appearance according to the viewer's relationship in space to the sculpture, the sound sculpture will also change it's sonic appearance as the observer moves in the space.

I created the sound sculpture specifically for the stairwell as it is made out of concrete which is highly reverberant and would not absorb the sound waves. The stairwell is also of a unique shape which makes the relationship and the activity of the standing waves within the space more complicated and unpredictable.

This paper will discuss the physical reasoning as to how this sound sculpture can exist, what technology and methodology I used in creating the sound sculpture including computer programs.

I will cite previous examples of this type of sound sculpture in order to show similarities and differences between my work and that of other composers.

Biography

Russell Goodwin is a Melbourne based composer working predominantly with electronic and concrete sound sources. He has created music for installations, film and theatre. Graduating with first class honours in composition from Monash University in 2000, he is currently researching the work of John Cage at Monash University.